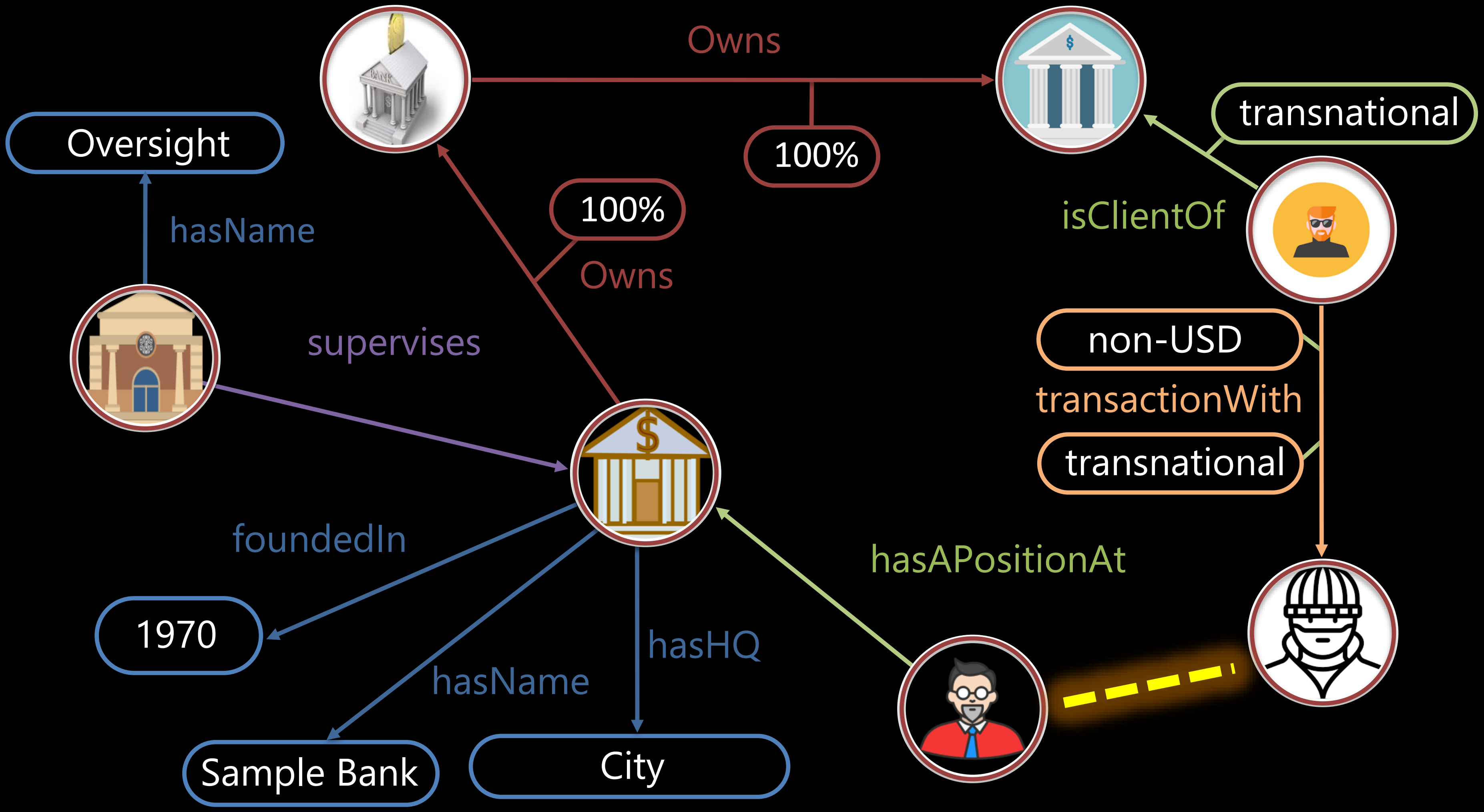
Knowledge Graph Embeddings
TransE

Emanuel Sallinger


Owns
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non-USD
transactionWith
transnational
foundedIn
hasAPositionAt
1970
hasHQ
hasName
Sample Bank
City

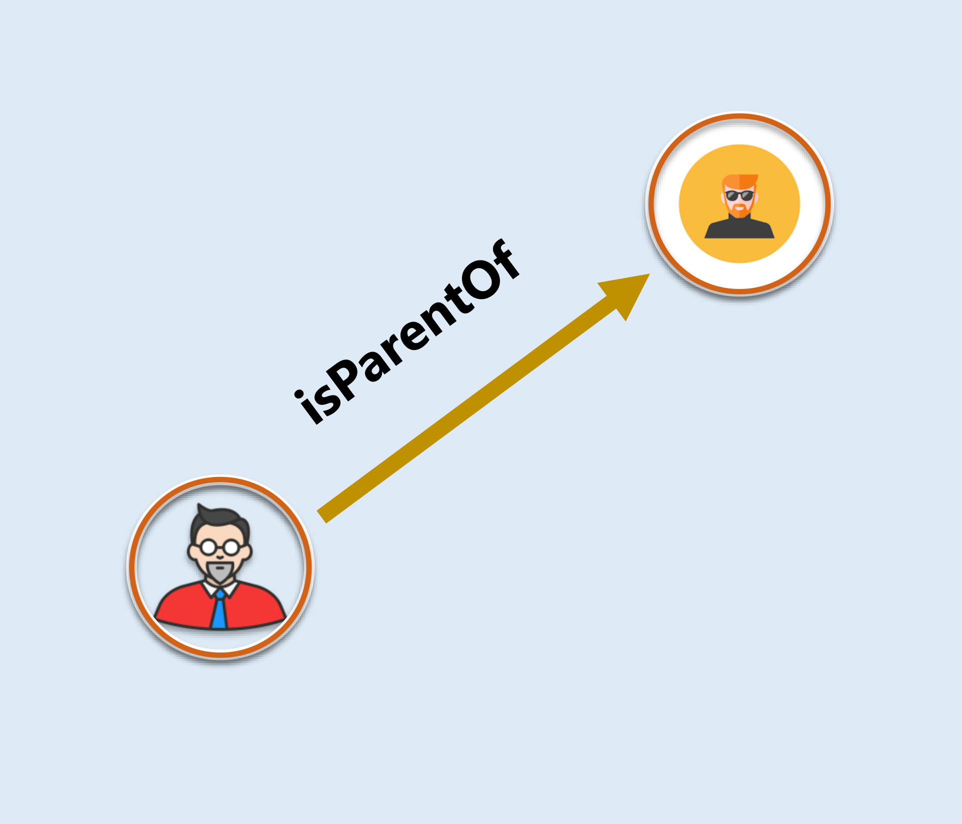
Main Parts

1. **Score** Function
   how likely does a particular edge hold?

2. **Loss** Function
   what is our objective for optimization?

3. **Learning** Algorithm
   integrating the loss and score function

$$e\colon (\mathbb{V} \cup \mathbb{E}) \to \mathbb{R}^d$$



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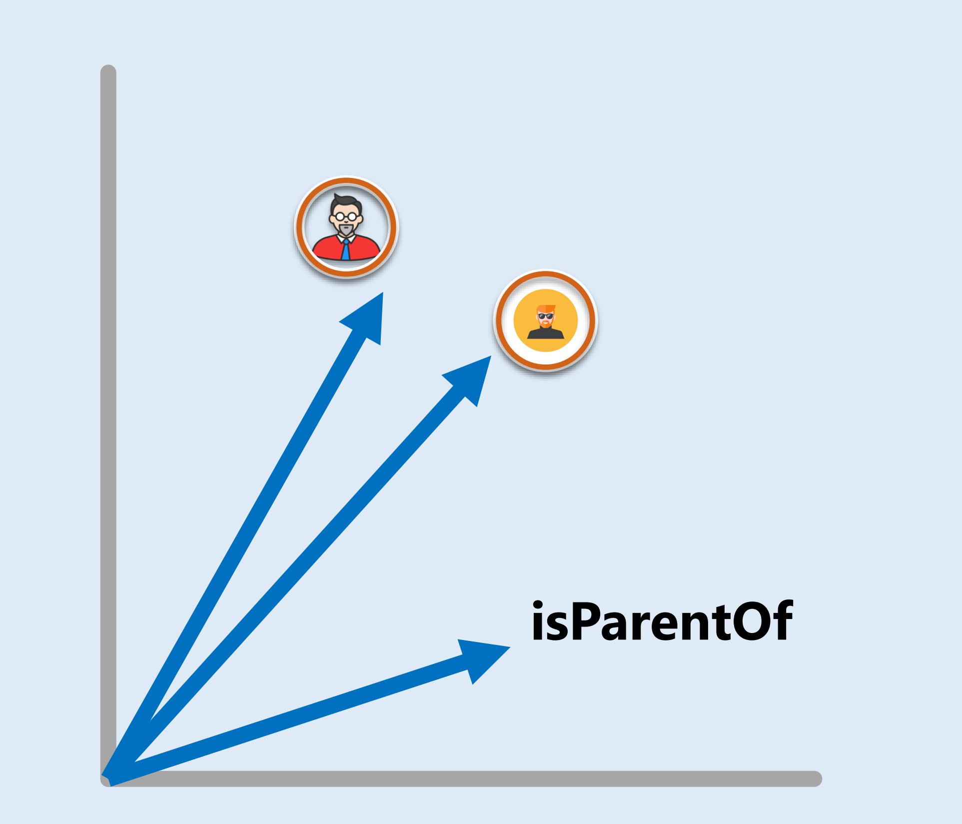

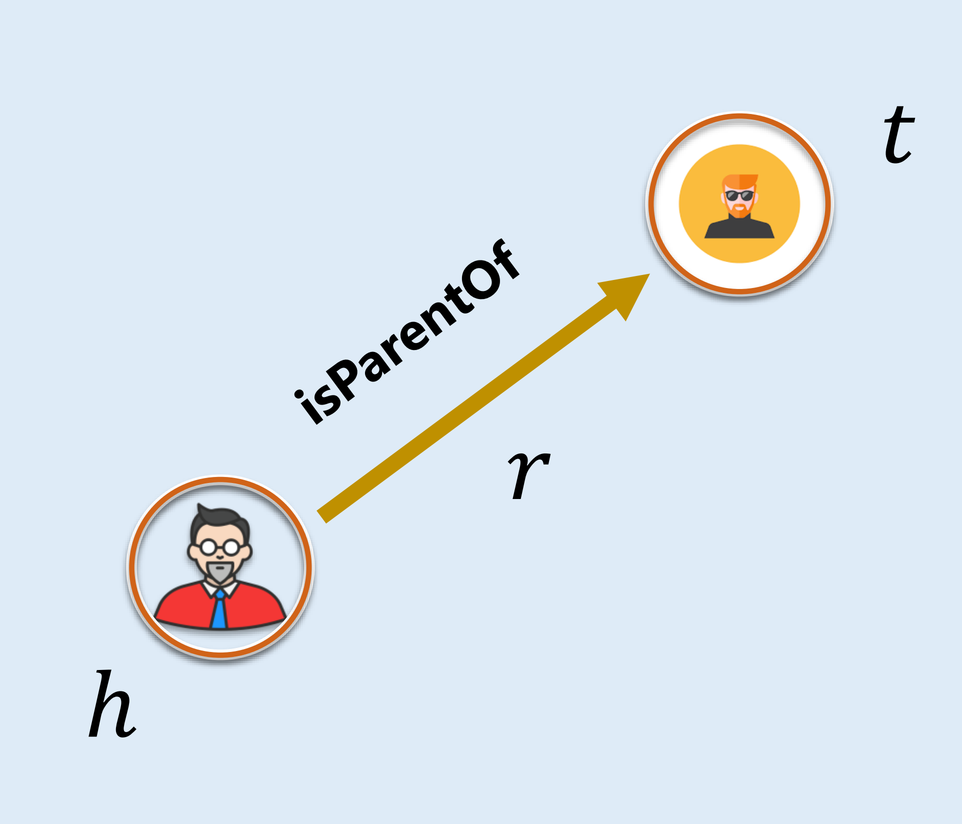

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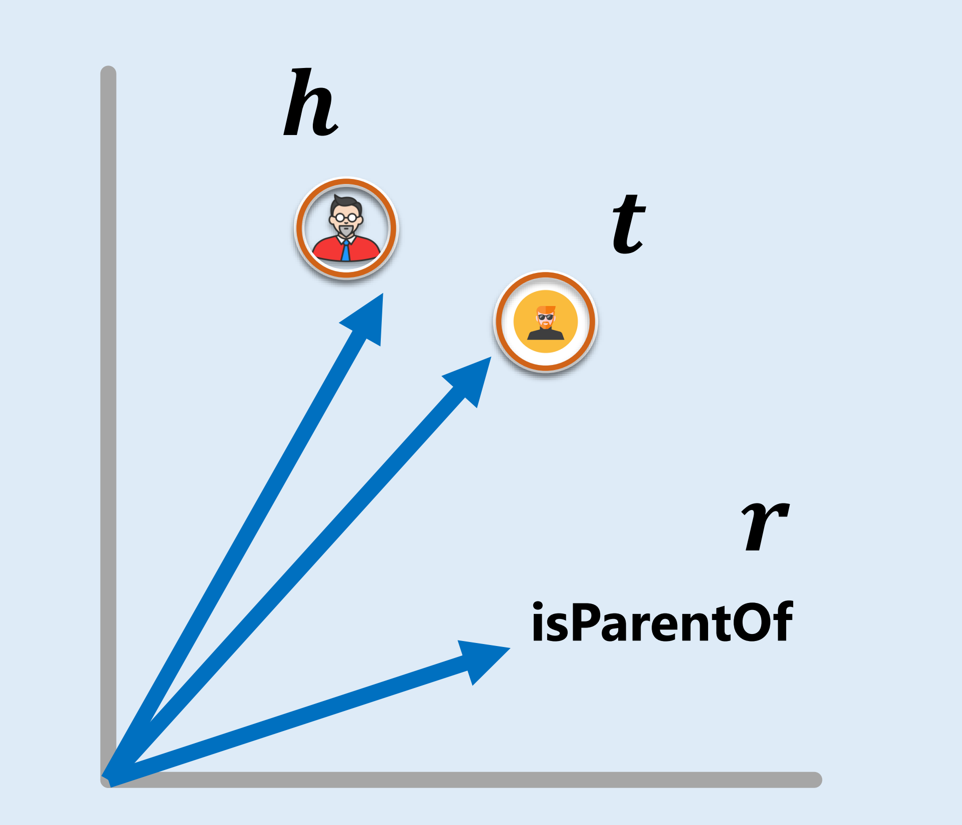

h
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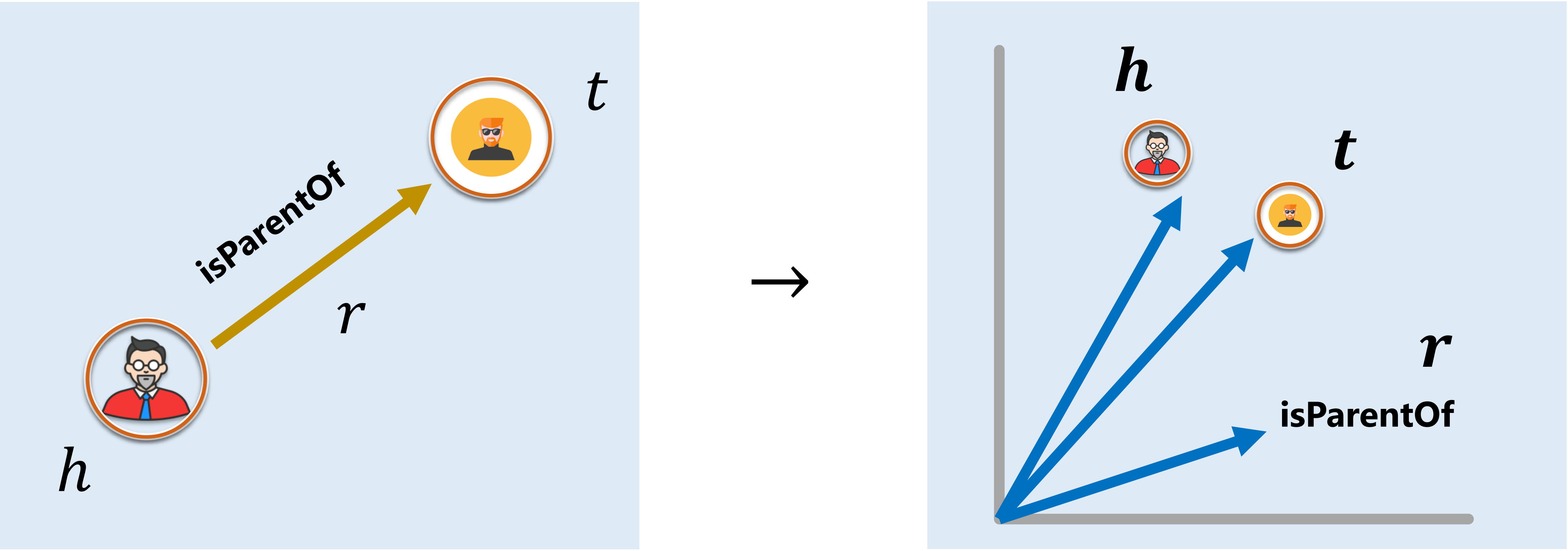
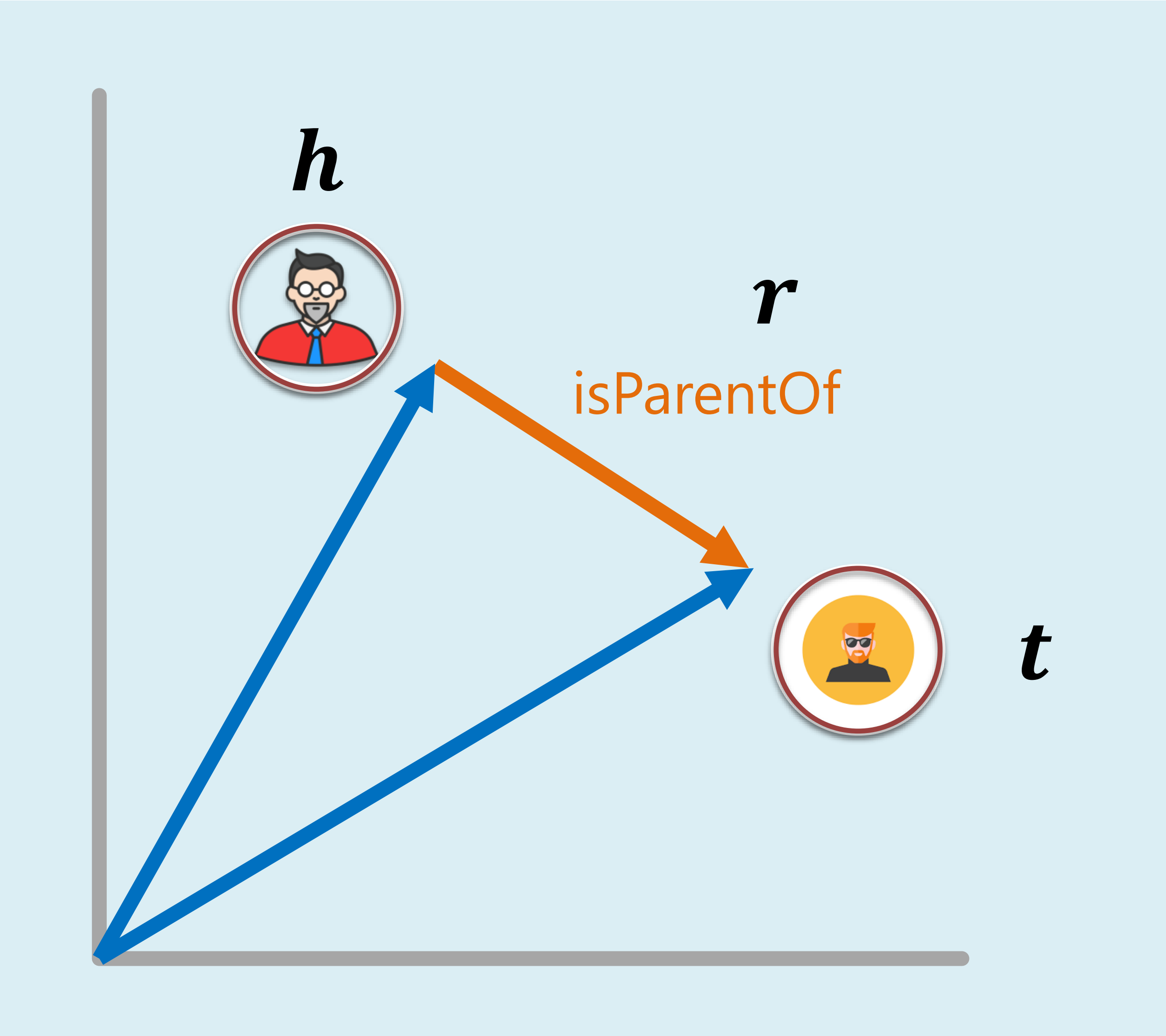
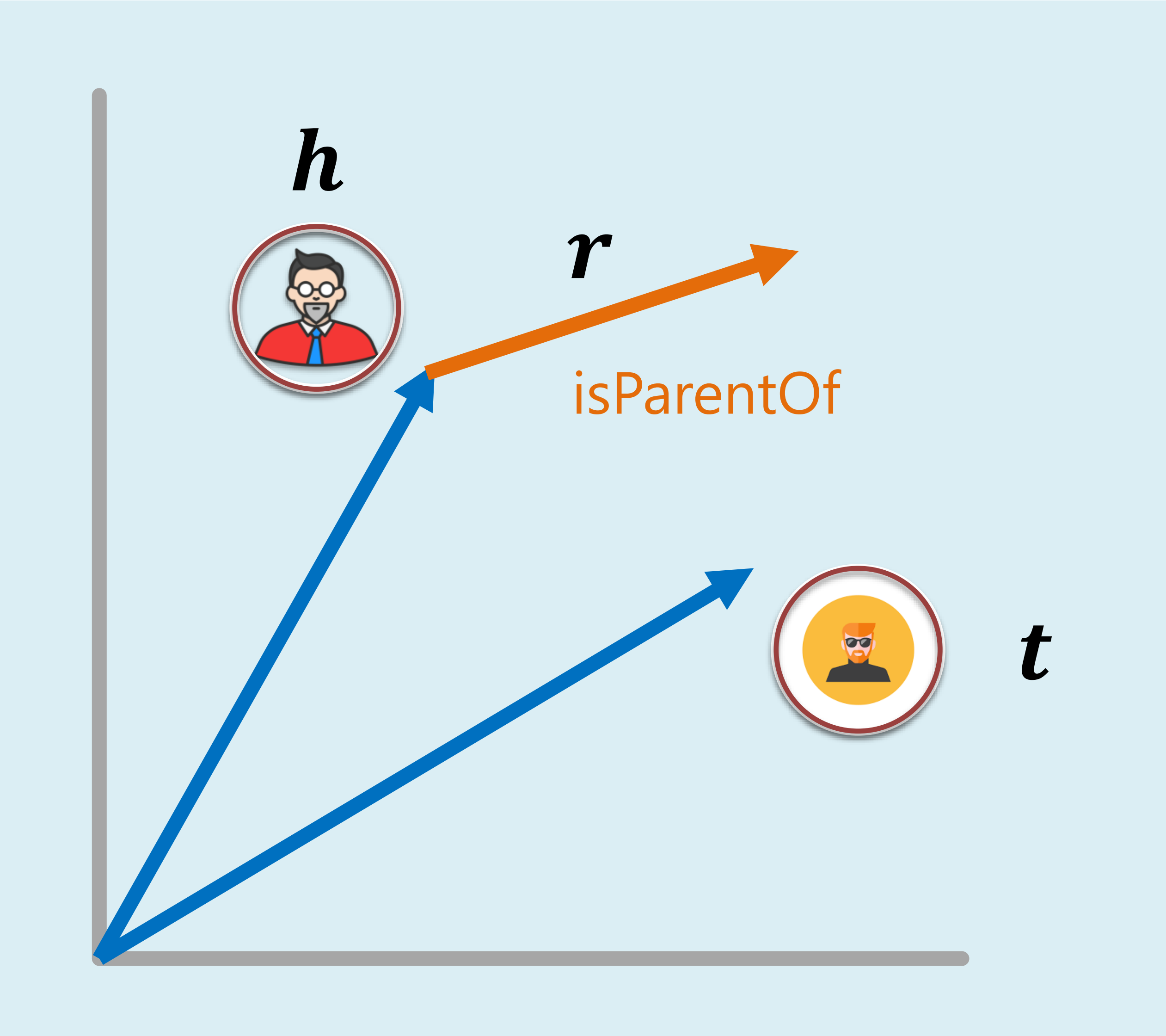
$$e(h) = \boldsymbol{h} \quad e(r) = \boldsymbol{r} \quad e(t) = \boldsymbol{t}$$





$$\boldsymbol{h} + \boldsymbol{r} \approx \boldsymbol{t}$$



$$\boldsymbol{h} + \boldsymbol{r} \approx \boldsymbol{t}$$

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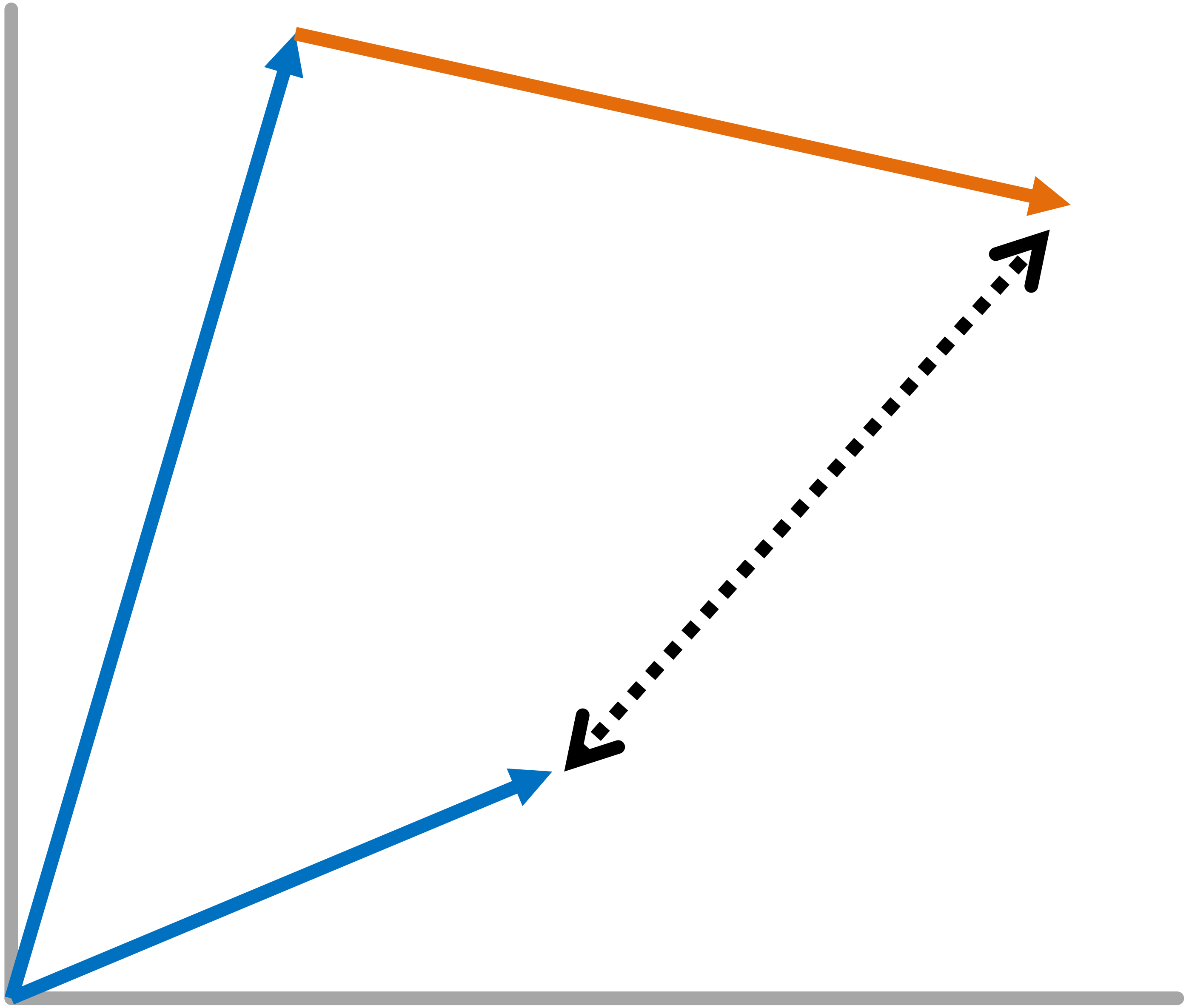
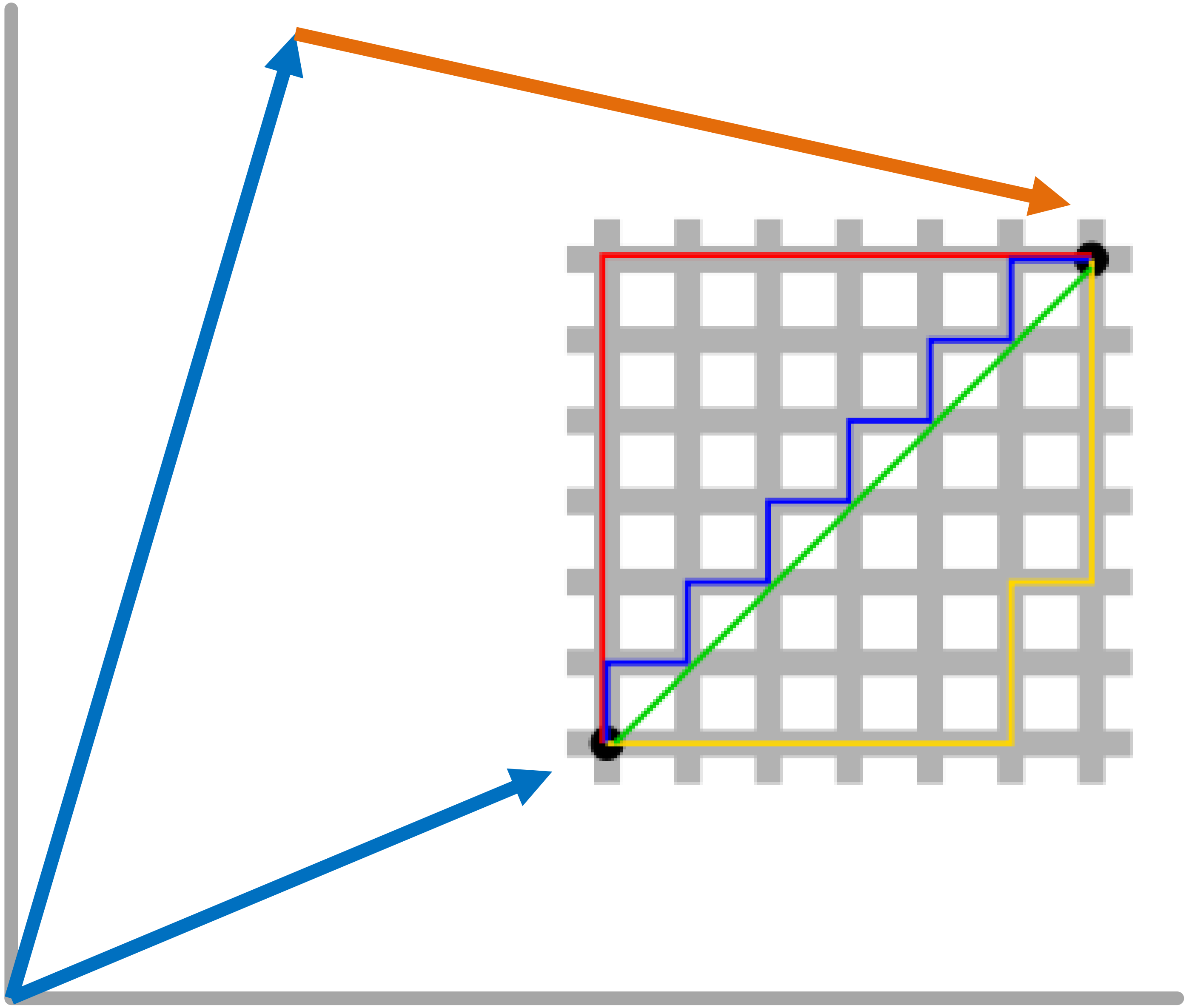


Euclidian distance ("L2-norm")

$$\|\boldsymbol{x}\|_2 = \sqrt{\sum_{i=1}^{d} |\boldsymbol{x}_i|^2}$$



“Manhattan” distance (“L1-norm”)

$$\|\boldsymbol{x}\|_1 = \sum_{i=1}^{d} |\boldsymbol{x}_i|$$

Source: https://en.wikipedia.org/wiki/Taxicab_geometry#/media/File:Manhattan_distance.svg

Score Function

Intuition:

$$\boldsymbol{h} + \boldsymbol{r} \approx \boldsymbol{t}$$

Score Function:

$$f(\boldsymbol{h}, \boldsymbol{r}, \boldsymbol{t}) = \|\boldsymbol{h} + \boldsymbol{r} - \boldsymbol{t}\|_1$$

or L2 norm instead of L1 norm

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Knowledge Graph Embeddings:

Learning **Latent** Knowledge

$$KG = (D, K; g, l, r)$$

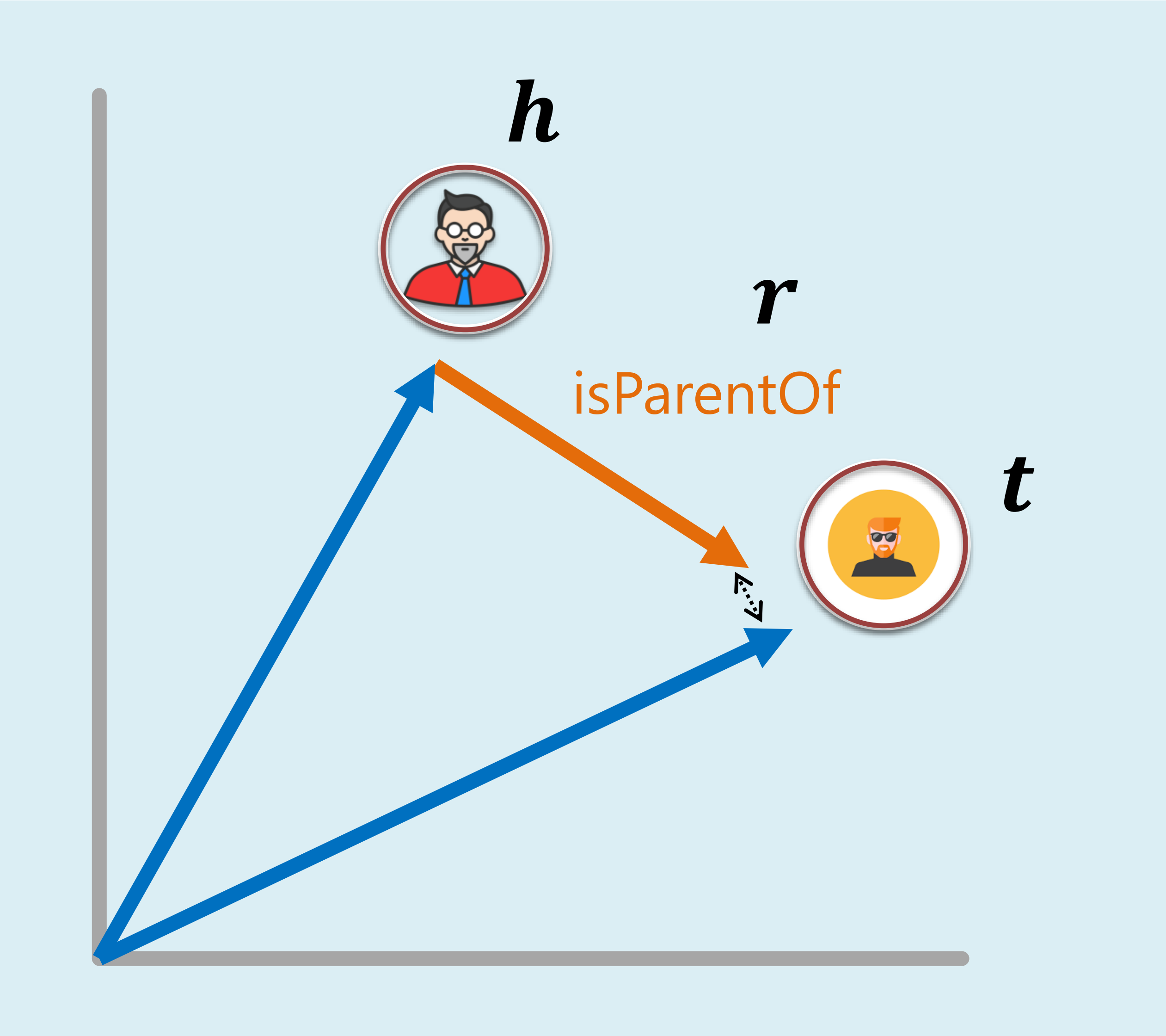
$$l: \mathbb{D} \rightarrow \mathbb{e}$$

Knowledge Graph Embeddings:

Reasoning with **Latent** Knowledge

$$KG = (D, K; g, l, r)$$

$$r_{\mathbb{e}}: \mathbb{D} \to \mathbb{D}$$


h
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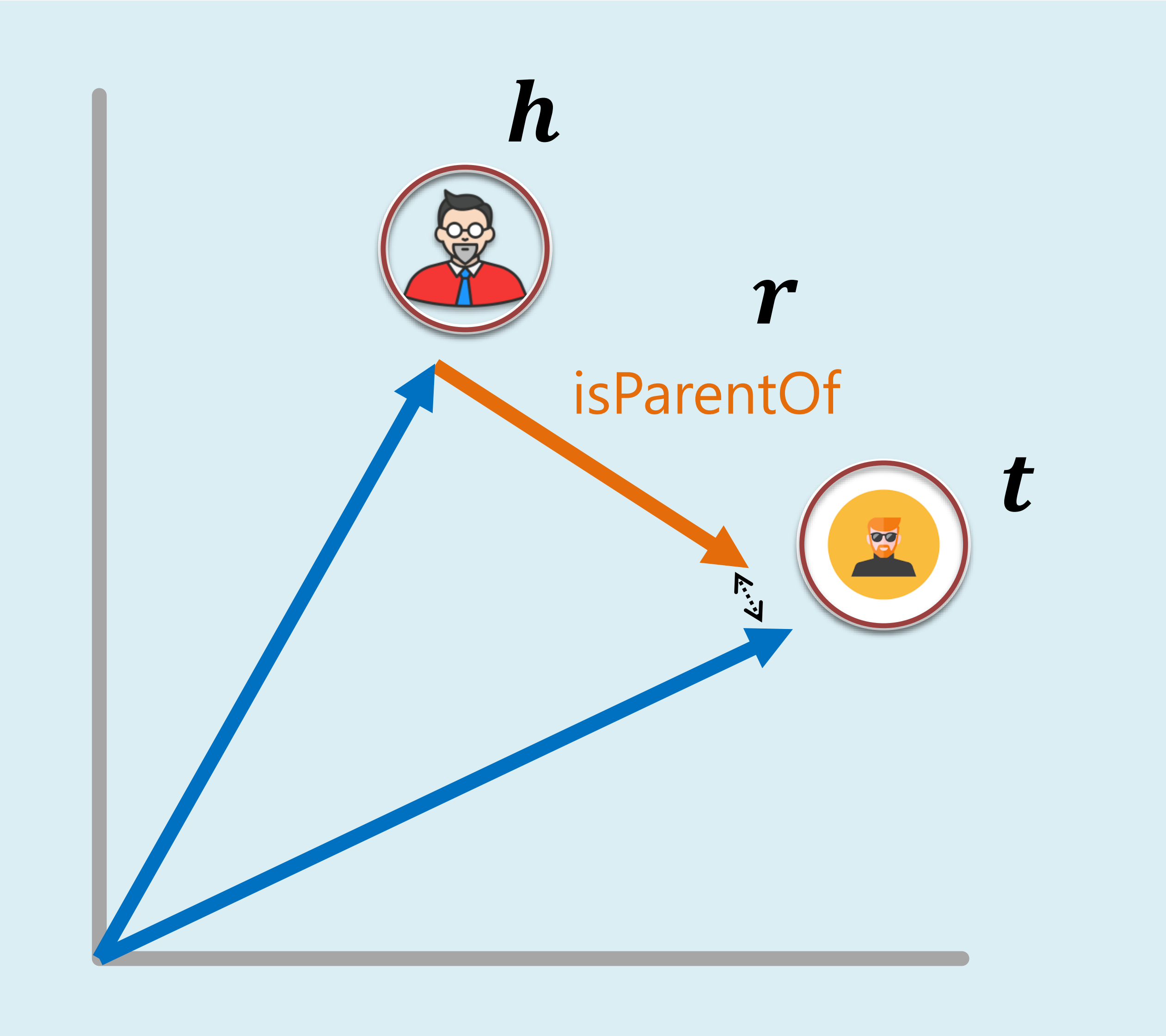
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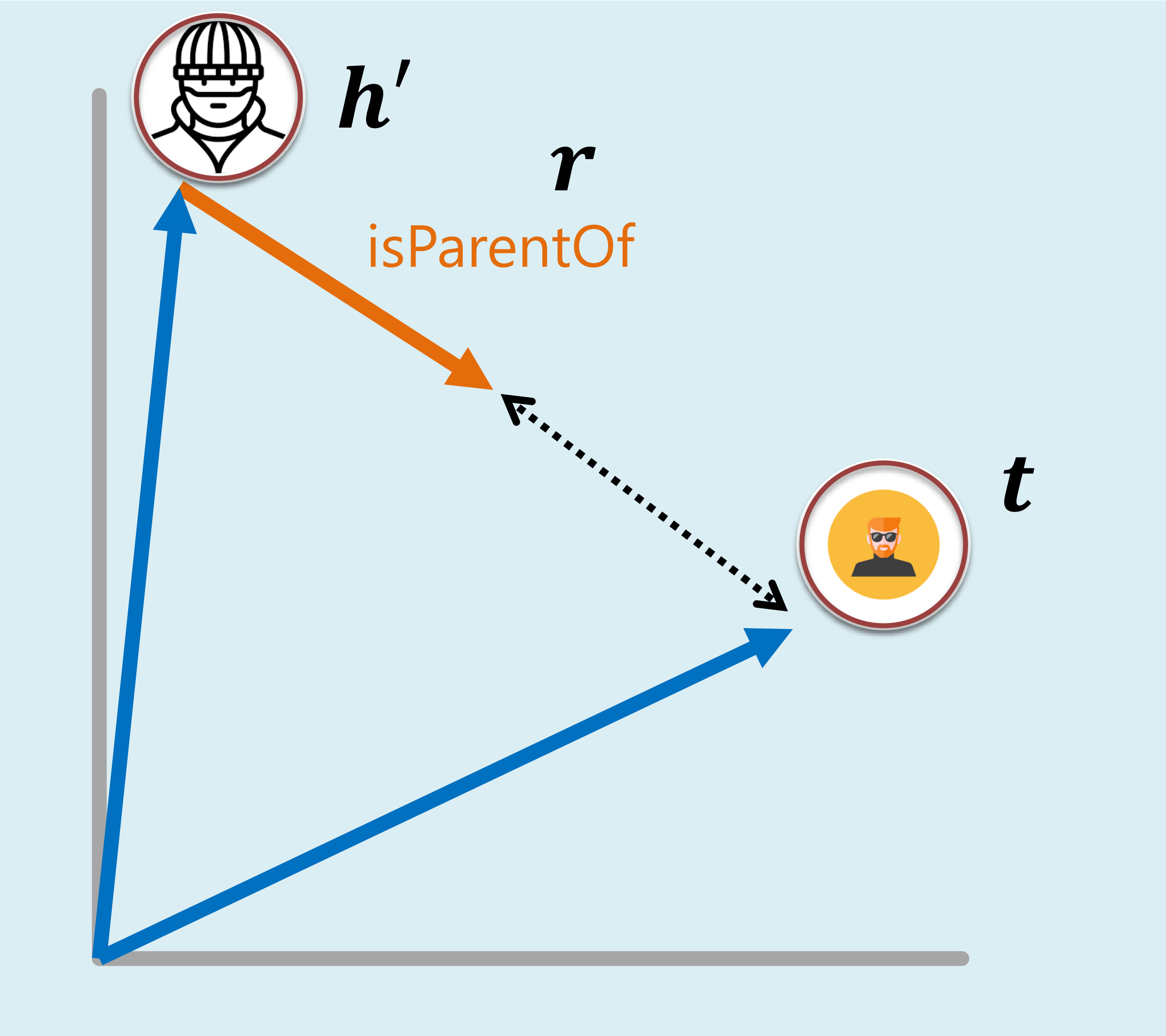
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